Reinforcement Learning for Dynamic Operation of a CO_2-to-Methanol Plant

Technical Report – extension of the EURECHA Process Design Contest

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Abstract

This report studies whether reinforcement learning can operate a cement-integrated CO_2-to-methanol plant profitably under volatile electricity prices. A Gymnasium environment couples hourly plant dynamics, economic rewards, GB/NL electricity-price traces, and Gaussian Process surrogate models derived from the Aspen-calibrated EURECHA base case. Soft Actor-Critic (SAC) is the strongest algorithm in the tested setup: under the internally consistent corrected v2 surrogate evaluation it achieves **2915.6** scaled £/week on GB prices, a **+487%** improvement over constant full-load operation, with 0% GPR extrapolation warnings. Cross-market transfer is promising: a GB-trained SAC evaluated on NL prices retains essentially the same original-v1 reward (3227 to 3222), although those absolute transfer values inherit the v1 extrapolation caveat.

The central limitation is surrogate trustworthiness. The original SAC policy triggered GPR extrapolation warnings on 96.1% of GB evaluation steps. A v2 pseudo-labelled surrogate correction removes that warning and preserves large RL value, but it has not been checked against Aspen in the low-load region. A v3 refit extends F_{H_2} coverage and fits cleanly ($R^2_{\mathrm{LCOM}} = 0.99981$), yet the v2-trained Fix (c) policy becomes uneconomic when scored on v3 (-161.6 ± 6.4), showing that fresh SAC retraining and Aspen low-load validation are still required. A flowsheet GNN extension is useful when reframed as per-edge physical-state prediction ($R^2 = 0.997$ across 104 outputs), not as direct graph-level LCOM prediction. PPO fixes, GNN confidence penalties, and the Pareto sweeps are kept as concise negative results.

1 Introduction

The parent EURECHA project designs a large-scale CO_2-to-methanol process co-located with a cement kiln. Fresh CO_2 (899 kmol/hr) is hydrogenated with green H_2 from a 270 MW PEM electrolyser. A recycle loop with purge achieves high overall conversion, producing approximately 213 kt/yr of methanol. The steady-state design and economic model are Aspen-calibrated at the base case and then represented by low-dimensional surrogate models for optimisation.

This extension asks a dynamic operating question: can an RL controller exploit hourly electricity-price volatility by changing electrolyser load and reactor conditions, instead of running the plant at constant full load? That question is economically important because electricity dominates variable cost for green methanol, and volatile day-ahead prices create hours where hydrogen production is either attractive or damaging.

The answer depends on two things. First, the policy must actually learn a profitable dispatch strategy. Second, the surrogate models used to score that strategy must be trustworthy in the operating region discovered by the agent. The main scientific story is therefore not simply "SAC wins". It is: *SAC can learn high-value dynamic operation, but surrogate coverage and pending Aspen validation are the limiting evidence gap.*

2 Methods

2.1 Gymnasium Environment

The environment is an hourly-step Gymnasium model with an 8-dimensional observation vector: electricity price, electrolyser load, H_2 buffer level, reactor temperature, reactor pressure, methanol rate, hour of day, and day of week. The continuous action vector sets electrolyser load, reactor temperature, and reactor pressure. First-order lags ($\tau_{\text{load}} = 0.3$, $\tau_T = 0.5$, $\tau_P = 0.7$ hr) give the plant inertia rather than letting actions jump the process state instantly.

2.2 Reward Function

The base reward is a scaled hourly economic profit:

$$R = 10^{-3}(\text{MeOH revenue} + \text{CO}_2 \text{ credits} - \text{electricity cost} \\ - \text{utility cost} - \text{ramp penalty} - \text{constraint penalties}). \tag{1}$$

Economic constants are single-sourced from the EURECHA project: e-methanol £800/t, CO_2 credits £80/t, 50 kWh/kg H_2 electrolyser specific power, and the Aspen heat-integration utility basis.

2.3 Surrogate Modelling

Four evidence tiers are kept separate throughout the report:

1. **Aspen-calibrated base case:** the original EURECHA steady-state process design and economics.
2. **Pseudo-labelled surrogate extensions:** v2 and v3 GPRs fitted to the simplified process model, not to new Aspen runs.
3. **Corrected surrogate evaluation:** SAC policies evaluated on internally consistent v2/v3 surrogate surfaces with explicit extrapolation monitoring.
4. **Pending Aspen validation:** 20 low-load validation points have been generated but not yet run in Aspen Plus.

The original v1 GPRs are trained on a 500-point Latin Hypercube sample from the simplified process model around the Aspen-calibrated design envelope. The v2 correction augments that set with pseudo-labelled points in the SAC operating region. The v3 correction extends the LCOM/utility F_{H_2} domain to [800, 3200] kmol/hr and is evaluated as a follow-up pseudo-labelled surface, not as an Aspen-labelled surface.

2.4 Agents and Baselines

Three agents are tested: SAC, PPO, and tabular Q-learning. SAC and PPO use Stable-Baselines3 continuous-control policies; Q-learning uses a $10 \times 5 \times 5$ discretised action grid. Baselines are constant full-load operation at nominal reactor conditions and a price-threshold rule that reduces load during high-price hours.

3 Base RL Results

Table 1 is the most important RL result because it places the original v1 performance beside the corrected v2 evaluation. The original SAC reward is high, but it is not a defensible standalone profit estimate: it triggers extrapolation warnings on 96.1% of GB evaluation steps. The current best internally consistent corrected v2 policy is Fix (c), the v2 surrogate plus variance-penalty SAC.

Table 1: GB evaluation summary from available outputs. Rewards are scaled £/week. v1 rows are useful for algorithm ranking but carry the GPR-extrapolation caveat; Fix (c) is the corrected v2 result.

Policy	**Reward**	**Std**	**CO_2 util.**	**GPR extrap.**
Full-load baseline	496.7	821.2	0.938	0.0%
Rule-based threshold	1764.9	177.2	0.943	0.0%
Original PPO (v1)	1710.2	425.1	0.935	97.0%
Q-learning (v1)	2700.5	214.7	0.935	37.2%
Original SAC (v1)	3194.4	52.1	0.935	96.1%
Fix (c) SAC (v2 + pen.)	**2915.6**	**35.8**	0.848	**0.0%**

The corrected v2 result is still large: Fix (c) improves on constant full-load by +487% and on the rule-based threshold by about 65%. This is the defensible dynamic-dispatch contribution. SAC is also the strongest algorithm in the tested setup, while PPO is retained mainly as a collapsed algorithmic baseline and Q-learning as a coarse but informative comparison.

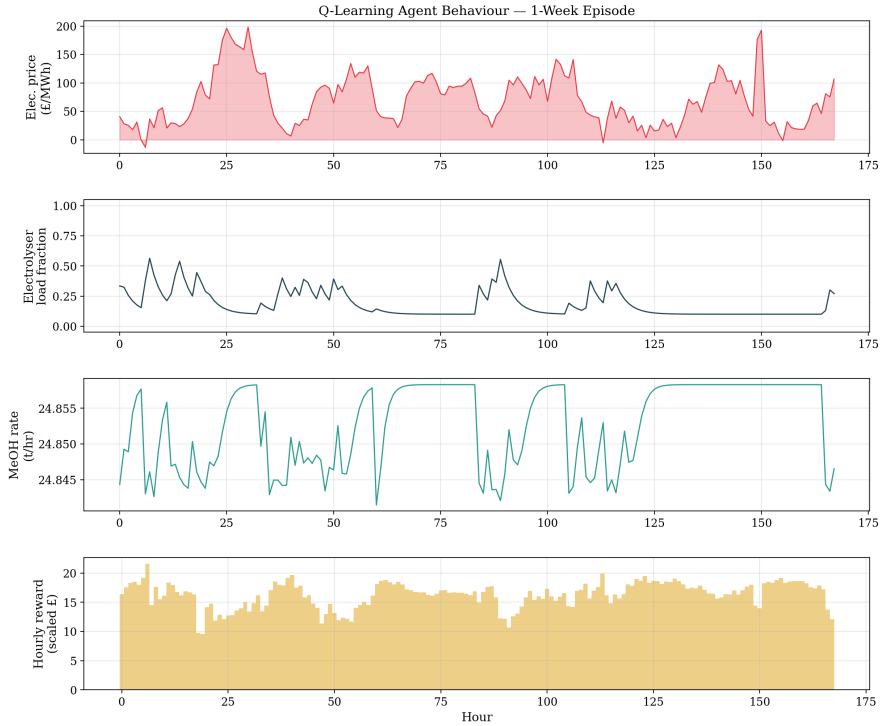


Figure 1: Representative learned dispatch behaviour over a 168-hour episode. The qualitative control pattern is curtailment during high-price hours and higher operation during cheap hours.

4 Cross-Market Transfer

The price loader supports GB synthetic prices and Dutch ENTSO-E-style 2023–2024 price data. The transfer experiment trains SAC, PPO, and Q-learning on GB or NL and evaluates each agent on both markets. These are original-surrogate transfer tests, so their absolute reward values inherit the v1 extrapolation caveat. The ranking and transfer pattern are still useful: SAC is the strongest transferred policy, and reward changes little when moving GB-trained SAC to NL prices.

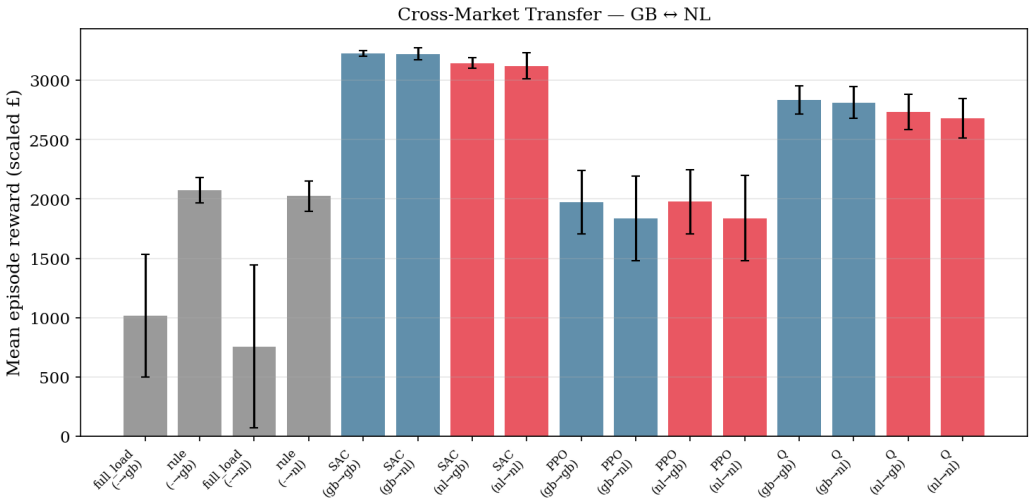


Figure 2: Cross-market transfer on the original surrogate setup. SAC transfers well across GB and NL price traces, but the absolute values should be read with the v1 extrapolation caveat.

Table 2: Cross-market transfer matrix (mean weekly reward ± std).

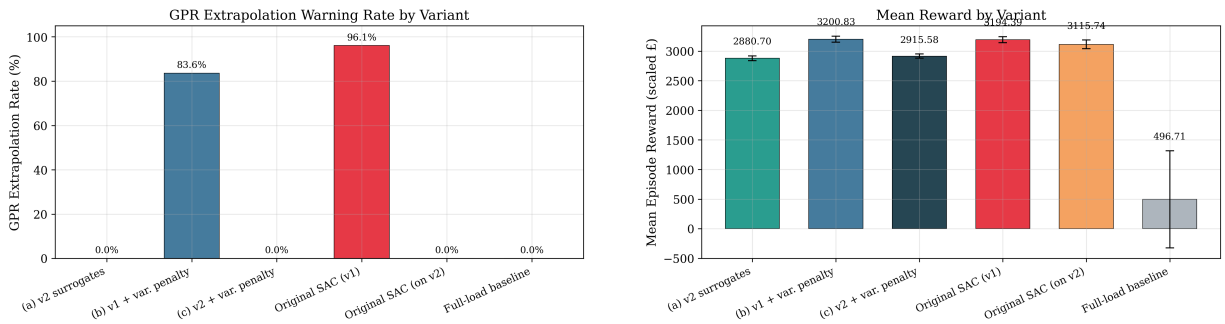
Agent	**GB→GB**	**GB→NL**	**NL→GB**	**NL→NL**
SAC	3227 ± 26	3222 ± 50	3146 ± 42	3123 ± 112
PPO	1972 ± 267	1838 ± 355	1976 ± 273	1839 ± 358
Q-learning	2836 ± 120	2814 ± 134	2735 ± 149	2682 ± 167

The GB-trained SAC changes from 3227 on GB to 3222 on NL, approximately 0.1% degradation. The NL-trained SAC also remains within 3% when evaluated on GB. A temporal robustness study on NL folds points in the same direction: the GB-trained zero-shot SAC scores 3180–3287 across held-out folds, while fold-retrained SAC ranges from 2328–3198. This supports the practical idea that a policy trained on a broad price distribution can transfer across nearby electricity markets, pending the surrogate validation caveat.

5 Surrogate Extrapolation Diagnosis and Correction

5.1 Original Problem

The environment flags GPR extrapolation when the MeOH surrogate predictive standard deviation exceeds twice the mean training-set standard deviation. The original SAC triggers this warning on 96.1% of GB evaluation steps. The agent has converged to a low-load, high-temperature, high-pressure region that the original 500-point LHS did not cover. Therefore, the original SAC reward is best interpreted as an optimistic pre-correction diagnostic, not as an Aspen-checked profit estimate.



(a) Warning rate.

(b) Episode reward.

Figure 3: Extrapolation-fix comparison. v2 surrogate coverage removes the warning; the v1 variance penalty alone does not.

5.2 v2 Correction

The v2 correction retrains surrogate models on the original 500 LHS points plus 300 pseudo-labelled points biased toward the agent trajectory. The labels come from the simplified process model, not Aspen. Approach B adds a variance penalty, but the main correction is the wider v2 surrogate coverage.

Table 3: Extrapolation-fix variants. Source: `extrapolation_fix_results.csv`.

Variant	Surr.	Penalty	Reward	Std	CO_2 util.	Extrap.
Original SAC	v1	No	3194.4	52.1	0.935	96.1%
Original on v2	v2	No	3115.7	74.2	0.916	0.0%
Fix (a)	v2	No	2880.7	39.8	0.836	0.0%
Fix (b)	v1	Yes	3200.8	50.6	0.935	83.6%
Fix (c)	v2	Yes	2915.6	35.8	0.848	0.0%
Full-load baseline	v1	No	496.7	821.2	0.938	0.0%

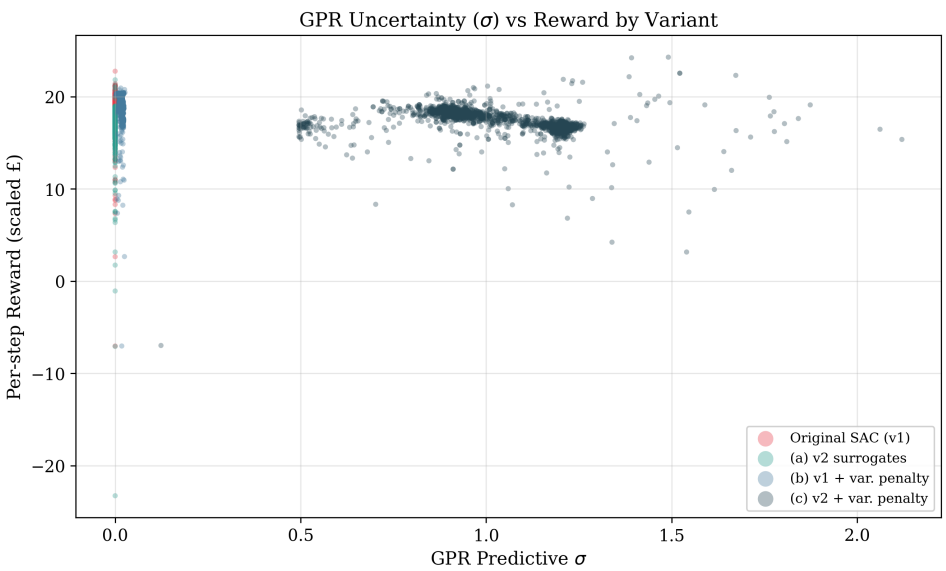


Figure 4: GPR predictive σ versus per-step reward. The original SAC occupies high-σ regions; v2 policies move into the corrected surrogate coverage.

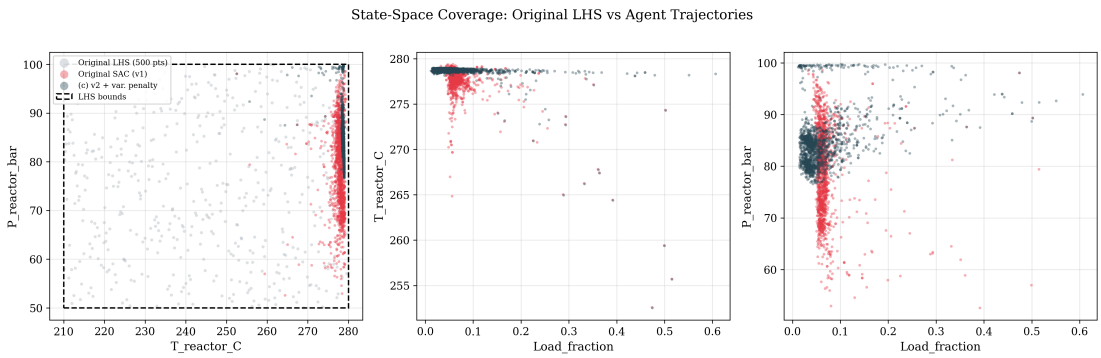


Figure 5: State-space coverage in T–P, load–T, and load–P projections. The discovered SAC operating region lies away from the original LHS; v2 is a pseudo-labelled attempt to cover that region.

The honest interpretation is that v2 gives an internally consistent corrected evaluation. It removes the warning and retains large RL value, but it cannot by itself establish physical accuracy in the low-load region because the new labels were not produced by Aspen.

5.3 v3 Extended F_{H_2} Domain

GNN diagnostics exposed a second surrogate issue: the original 5-D LCOM GPR was fitted on $F_{H_2} \in [2515, 3147]$ kmol/hr, while the dynamic pipeline can map low PEM capacity factors to much lower F_{H_2} values. The v3 saved surrogates extend the fitted LCOM/utility domain using 500 additional pseudo-labelled LHS points with $F_{H_2} \in [800, 3200]$ kmol/hr, combined with the original 500 points.

Table 4: v3 held-out fit metrics from `outputs/surrogate_v3/summary.json`. These are surrogate fit metrics against pseudo-labels, not Aspen validation metrics.

GPR target	Test R^2	Test RMSE
LCOM	0.99981	0.98 £/t
Utility	0.99936	1.1×10^{-3}

When the v2-trained Fix (c) policy is scored on v3 without retraining, mean reward collapses to -161.6 ± 6.4 with 0% extrapolation. This is not a failure of SAC as an algorithm; it shows that the low-load F_{H_2} domain is economically important and that the previous v2 policy was not trained on the latest intended v3 surface. The v3 result requires fresh SAC retraining and Aspen low-load spot checks before updated economic claims can be made.

6 GNN Flowsheet Surrogate Extension

6.1 Graph-Level KPI Failure

A topology-aware GINE model was first tested as a direct graph-to-KPI surrogate for [TAC, carbon efficiency, LCOM]. The best graph-level configuration fits TAC and carbon efficiency reasonably well but fails on LCOM. This is an informative ablation: LCOM is a low-dimensional position-specific GPR distillation, and graph-level mean/add pooling discards the positional information required to reconstruct it.

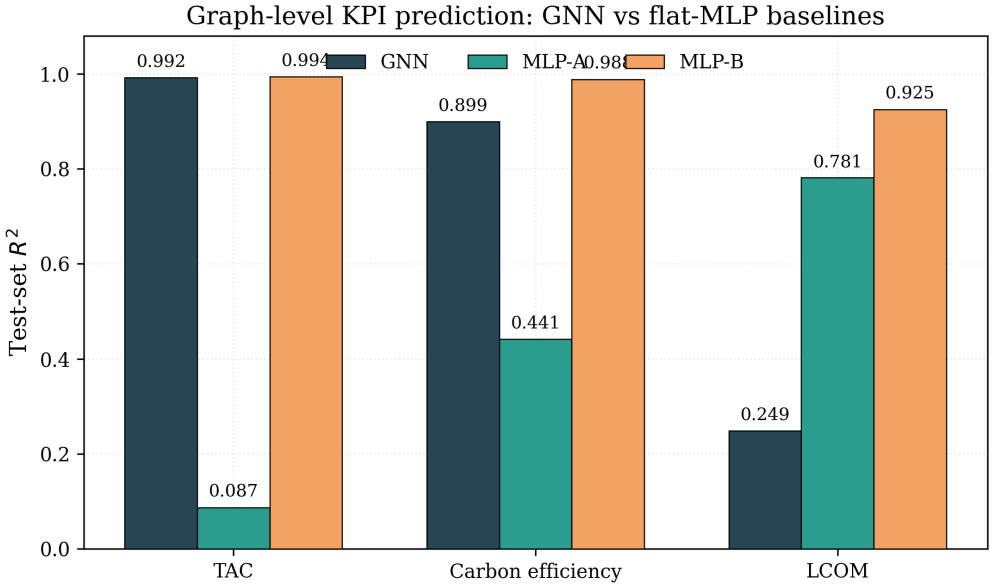


Figure 6: Graph-level KPI prediction. The GNN under-fits LCOM; flat MLP baselines reveal that this is a task-formulation mismatch, not just a lack of learnable signal.

Table 5: Graph-level KPI regression and flat baselines. The edge-augmented MLP result is not treated as a production model because edge mass flows leak production information into the input.

Model	$\mathbf{R^2_{TAC}}$	$\mathbf{R^2_{carbon}}$	$\mathbf{R^2_{LCOM}}$
GNN (best graph-level config)	0.992	0.899	0.249
MLP-A (design only)	0.087	0.441	0.781
MLP-B (design + edge-flat)	0.994	0.988	0.925
Cap-factor fix GNN	0.991	0.894	0.375

6.2 Edge-State Reframe

The successful reframe is per-edge state prediction. The model predicts eight physical quantities for each of 13 edges:

$$[\dot{m}, T, P, x_{\mathrm{CO_2}}, x_{\mathrm{H_2}}, x_{\mathrm{MeOH}}, x_{\mathrm{H_2O}}, x_{\mathrm{inert}}]\,.$$

The two structural flags (`is_recycle`, `is_energy`) remain as edge inputs, while physical edge quantities are moved to the target to avoid leakage. This aligns the learning task with graph message passing.

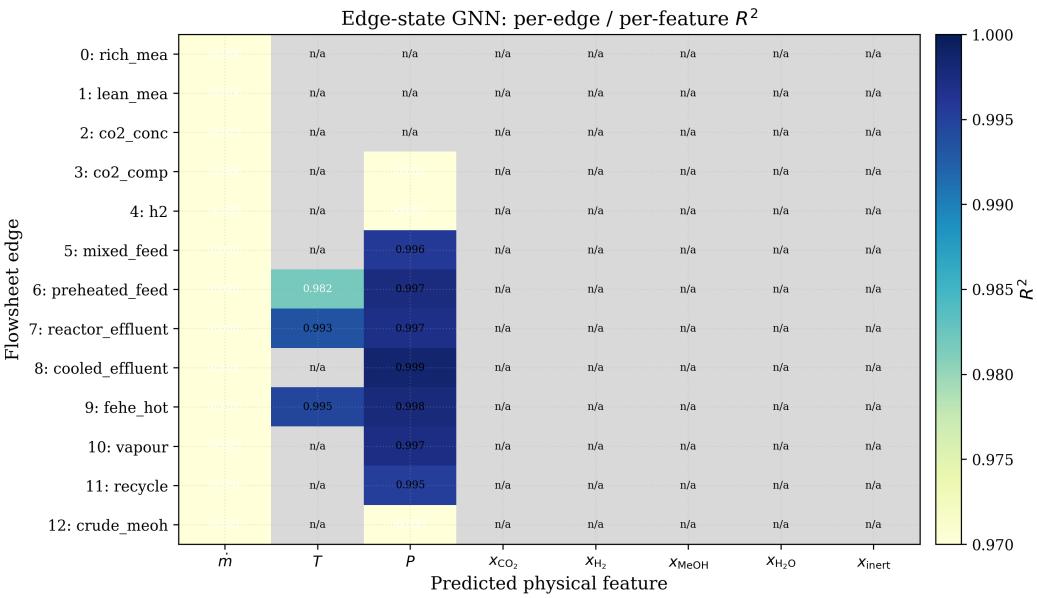


Figure 7: Per-edge/per-feature R^2 map for the edge-state GNN. Most informative physical channels are fitted at very high accuracy.

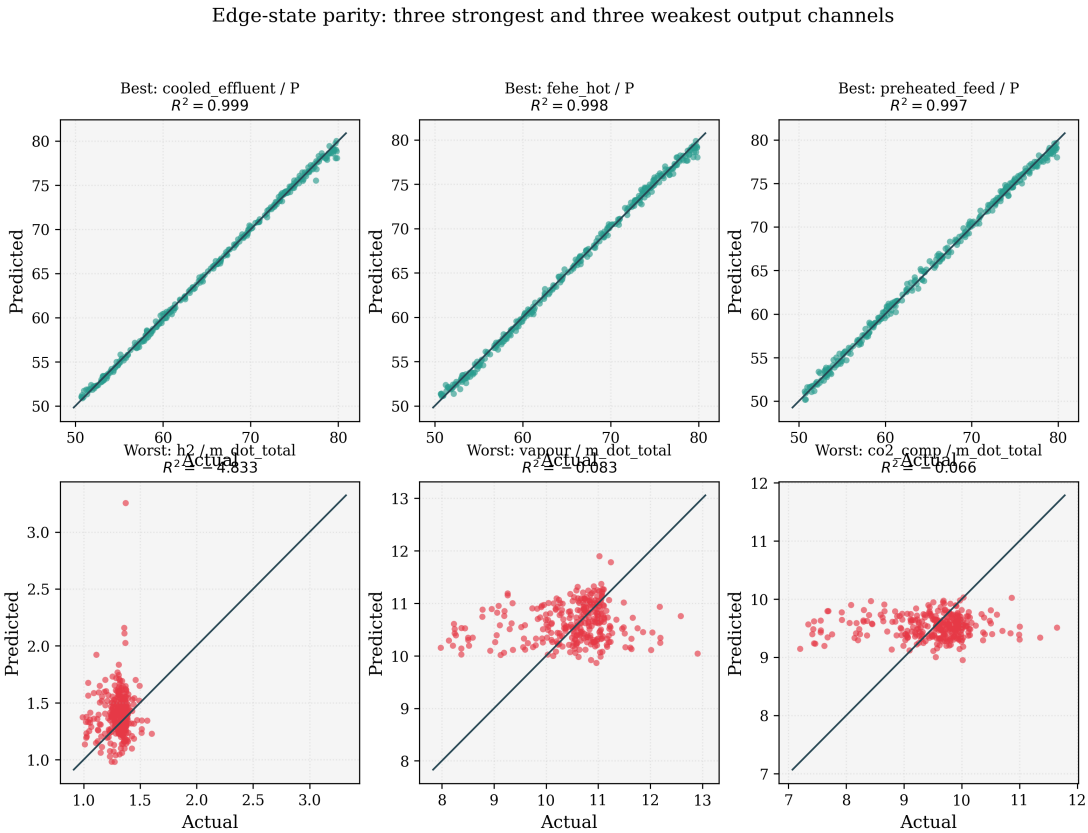


Figure 8: Parity plots for selected edge-state channels. The edge-state task avoids the direct graph-level LCOM collapse.

The result is strong for plant-state prediction: mean per-edge R^2 is 0.997, worst feature R^2 is 0.991 for total mass flow, and temperatures and compositions are near 0.9999. A derived methanol-rate proxy from the distillate edge remains weak ($R^2 \approx -0.06$), so the defensible architecture is not "GNN directly predicts all economics". It is: edge-state GNN for structured plant

state, scalar GPR heads for LCOM/utility, and explicit TAC formulas where available.

7 Additional Negative Experiments

These experiments are kept because they narrow the design space, but they do not drive the main claim.

7.1 PPO Fix Sweep

The original PPO transfer results suggested a constant-policy collapse. Four standard interventions were tested for 500k steps: larger rollout horizon, smaller learning rate, more epochs, and VecNormalize, individually and combined. Only the normalised variants develop nonzero load variance; all PPO variants keep reactor temperature and pressure variance at zero.

Table 6: PPO fix sweep from `outputs/ppo_fix_results.csv`. VecNormalize rows use a different reward scale from plain PPO rows.

Experiment	Algo	Reward	Std	var(load)	var(T)	var(P)
`sac_original`	SAC	1139.9	481.0	1.1×10^{-3}	430.9	249.7
`ppo_vecnormalize`	PPO	961.7	469.5	0.023	0.0	0.0
`ppo_fixed_combo`	PPO	934.2	457.1	0.013	0.0	0.0
`ppo_original`	PPO	413.4	484.9	0	0	0
`ppo_nsteps_4096`	PPO	413.4	484.9	0	0	0
`ppo_lr_1e4`	PPO	413.4	484.9	0	0	0
`ppo_epochs_20`	PPO	413.4	484.9	0	0	0

7.2 GNN Confidence Penalty

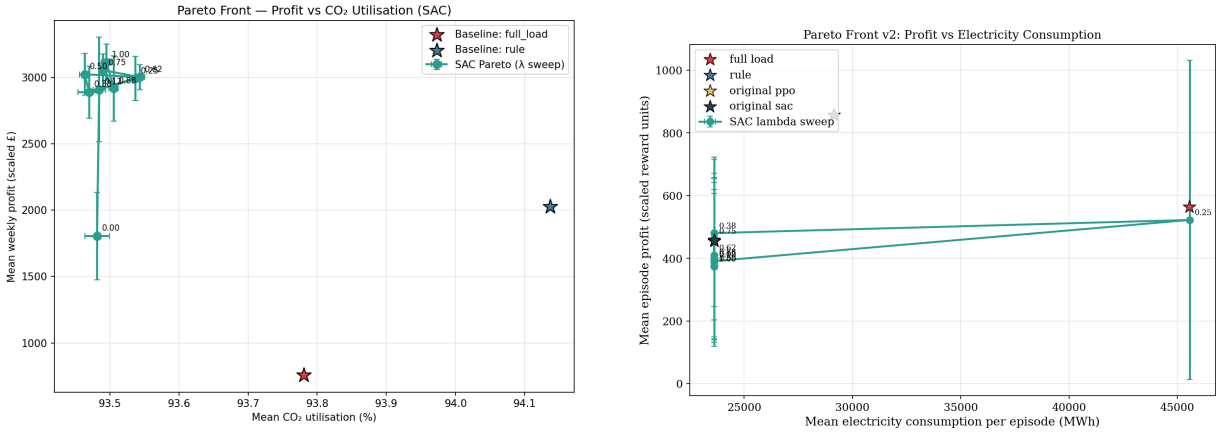
An edge-state GNN uncertainty penalty was added to the SAC reward on v2 surrogates. It underperforms Fix (c): mean reward falls from 2915.6 to 719.9 and variance rises sharply. The likely reason is that the v2 GPR coverage problem is already removed, so the GNN penalty becomes a broadly applied negative shaping term rather than a targeted novelty penalty.

Table 7: GNN-confidence penalty versus Fix (c).

Variant	Reward	Std	CO_2 util.	Extrap.	$\bar{\sigma}_{\text{GNN}}$
Fix (c) v2 + variance penalty	2915.6	35.8	0.848	0.0%	—
Edge-state GNN penalty	719.9	590.1	0.808	0.15%	0.183

7.3 Pareto Degeneracy

The profit/CO_2-utilisation Pareto sweep is degenerate: CO_2 utilisation stays near 93.5% across the sweep. The profit/electricity reframe is also mostly degenerate because electricity cost is already inside the profit term. Eight of nine electricity-weighted SAC runs collapse to approximately 23,615 MWh/episode; only one run jumps to a full-load-like 45,567 MWh/episode.



(a) Profit vs. CO_2 utilisation.

(b) Profit vs. electricity consumption.

Figure 9: Both Pareto formulations are effectively degenerate for the current reward structure.

8 Aspen Validation Gap

The remaining physical validation gap is explicit. The v2/v3 extensions are pseudo-labelled by the simplified process model; they are not new Aspen surfaces. To make the next validation step concrete, the repository includes `scripts/generate_aspen_validation_points.py`, which clusters the 3,360 rows of Fix (c) step data into 20 representative operating centroids and exports `outputs/aspen_validation/validation_points.csv` plus an `aspen_variables.txt` mapping.

Table 8: Summary of the 20 Fix (c) Aspen validation centroids. These points have been generated but not yet run in Aspen Plus.

Statistic	**load**	T **(°C)**	P **(bar)**	F_{H_2} **(kmol/hr)**
min	0.02	253.8	68.8	57
mean	0.15	275.3	87.7	403
max	0.45	278.9	99.2	1211

Sixteen of the twenty centroids have $F_{H_2} < 800$ kmol/hr, below the v3 lower bound. This is precisely why the 20 Aspen runs are the next required step. They would not fully validate every possible low-load state, but they would provide the first Aspen spot checks in the policy's actual operating region and allow v3 predictions to be compared with physical labels.

9 Conclusions

1. SAC is the strongest algorithm in the tested setup. Under the internally consistent corrected v2 evaluation, Fix (c) achieves 2915.6 scaled £/week on GB prices, +487% versus full-load, with 0% GPR extrapolation warnings.
2. Cross-market transfer is promising. The GB-trained SAC retains nearly the same original-v1 reward on NL prices, but those transfer rewards should be treated as transfer evidence, not settled profit values.
3. The original GPR extrapolation problem was real. Original SAC triggered warnings on 96.1% of GB steps. The v2 pseudo-labelled correction removes the warning and preserves large RL value, but remains an internally consistent surrogate evaluation rather than physical validation.
4. The v3 extended-F_{H_2} result changes the interpretation: scoring the v2-trained Fix (c) policy on v3 gives -161.6 ± 6.4. The low-load domain is economically important, so SAC must be

retrained directly on v3 before updated economic claims are made.

5. Aspen validation points have been generated but not run. The 20 centroids are the next required physical spot checks, especially because 16 of 20 fall below the v3 F_{H_2} lower bound.
6. The GNN is useful for edge-state prediction, not direct graph-level LCOM prediction. The defensible architecture is edge-state GNN plus scalar KPI heads and explicit economics.
7. PPO collapse, GNN confidence penalties, and the Pareto sweeps are informative negative results. They support the main story by showing which tempting routes did not improve the current formulation.